Spike Traces as Calibrated Predictive States for Reconstruction-Free World Models

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v0.7.10b headline result (read first). This release adds the **v0.7.10b OOD Path-C** (3 DMC sub-families, 6 splits, 12 model variants, 39 held-out envs = 468 cells), which is *the gating experiment for the working title*. The headline number is STJEWM $\rho \in [0.9676, 0.9986]$ in every split, while non-SNN baselines each fail at a distinct axis. The paper's structure has been restructured around this OOD finding: §7 (Cross-Environment Transfer) now leads with the OOD Path-C matrix in §7.6 and demotes the v0.7.8 within-suite pilot to a single subsection. The 4 collapse-robust metrics (divergence, responsiveness, event-alignment ρ, env-SR) are introduced in §6 and re-applied as the *only* axes that discriminate the families. The working title "generalisable world models" is now supported *within DMC sub-families*. The cross-benchmark-family axis (Pusht / LeWM reacher / Tworoom / Delayed POMDP) and the cross-modality axis (state $\rightarrow$ pixel) are deferred to a future paper that requires a raw-obs branch in STJEWM. v0.7.10b contribution. All numbers in the tables are unchanged; only the interpretation is.

Abstract

The headline. Across 6 OOD splits $\times$ 12 model variants $\times$ 39 held-out DMC envs = 468 cells, the calibrated event-driven predictive state transfers across DMC sub-families (F1 classic control / F2 locomotion / F3 sparse-POMDP) under 1-, 2-, and 3-family held-out splits. STJEWM 6 readouts hold $\rho \in [0.9676, 0.9986]$ in every split; LeWM over-reacts (`resp` 2.4–6.2), GRU under-fits (`resp` 0.10), MLP collapses (`resp` 0.0007). The failure mode is intrinsic to the model class, not to the env list. This *supports* the working title "generalisable world models" within DMC.

The contribution. Reconstruction-free world models compress observation–action streams into a latent that the planner reads on every decision step. The dominant design — a continuous recurrent hidden state, unconstrained during the model's forward pass — is attractive but ambiguous: it is unclear whether such a state is a *predictive state* or merely an unrestricted recurrent memory. We argue that the right question is not *which world model predicts more accurately* but *what kind of latent state is a learnable, generalisable, planner-friendly predictive state*. We introduce **ST-JEWM**, a pure-SNN reconstruction-free world model whose predictive latent is a *gated exponential trace over post-spike activations*, and we couple it to the **membrane-forbidden protocol**: the planner is forbidden from reading the continuous membrane potential. Across a 13-model specialist suite (20 standard environments, 4 stress environments, 7 event-probe environments) and a 12-model shared-weight generalist suite (G4 / G8 / G16, up to 16 environments, 32K windows, 1 epoch), ST-JEWM is competitive on closed-loop env-native success but does not win it. We also show that the latent cosine-success metric, when used alone, can be inflated by collapsed representations, motivating collapse-robust diagnostics (divergence-from-constant, responsiveness, event-probe AUROC, event-alignment ρ). Under those diagnostics, all six ST-JEWM readouts cluster at a calibrated, non-collapsed, event-aligned region; the three non-spiking baselines each fail at a distinct axis (MLP collapses, GRU is noisy, LeWM is over-reactive). The results argue that the relevant design choice for reconstruction-free world models is not *which* continuous representation to learn, but *what* the planner and predictor are allowed to read — and that such claims require collapse-robust diagnostics rather than latent cosine success alone.

1 Introduction

Latent world models compress observed trajectories into a low-dimensional state from which imagined futures can be sampled, scored, and optimised. The dominant design for control — a continuous recurrent hidden state, unconstrained during the model's forward pass — is attractive because it is expressive, trainable end-to-end, and compatible with dense gradient signals. Its expressiveness, however, papers over an interface question that the field rarely confronts explicitly: when a planner takes the next action, *what part of the model is allowed to read?* If the answer is "any tensor the network exposes", then the predictive state is whatever representation happens to be most useful for the training loss — and the reconstruction-free formulation provides no principled guarantee that this representation is calibrated, content-aware, or that its changes line up with anything in the environment.

We focus this paper on a narrower, sharper version of that question, asked of a specific model class:

> If a spiking dynamical system is allowed to maintain a continuous membrane potential internally, but a downstream predictor or planner is forbidden from reading it, can the bounded post-spike event history of that system still serve as a usable, non-trivial predictive state?

The membrane-forbidden protocol is not a performance trick. It is an interface constraint that asks whether event history is *sufficient* as a predictive state — without the loophole of letting the planner read the unconstrained continuous variable that the spike train is meant to replace. Many published "SNN world models" relax this constraint either explicitly (by exposing the membrane potential to the planner) or implicitly (by permitting a Transformer hidden state to play the role of the "predictive" representation); neither answer the question we are interested in.

A second difficulty is evaluation. Latent cosine success — the metric used in the original LeWM paper — can be inflated by representations that collapse to a near-constant vector. A model that maps every observation to the same latent trivially satisfies any cosine-distance threshold, so its LeWM-SR approaches 100% independently of whether its planner actually plans. Across 12 generalist checkpoints on the G4 / G8 / G16 suites we find a stateless MLP baseline achieves LeWM-SR $\approx$ 95.5% while its per-dim latent standard deviation is *0.0002* — three orders of magnitude below every other model. This makes LeWM-SR unsafe as a head-line metric, but does not make it useless: when paired with collapse-robust measures it becomes a useful *upper-bound* proxy on planning competence, after which one asks "how does the model achieve that latent geometry?".

We therefore propose a coupled intervention: a stricter predictive-state interface (the membrane-forbidden protocol) *and* a coupled collapse-robust diagnostic package (env-native success, divergence-from-constant, responsiveness, event-probe AUROC, event-alignment ρ). The package is built around one principle: a metric should be *unfoolable* by a constant latent.

ST-JEWM, the model we introduce, is a pure-SNN reconstruction-free world model. Its encoder, dynamics, and predictor are all MultiComp SNN cells; its predictive latent is a *gated exponential trace over post-spike activations*. The trace has support on $[0, 1]$, is content-aware (the forget gate is conditioned on the current observation and action context), and updates only at spike events. Because the planner reads the trace and not the membrane potential, the membrane-forbidden constraint is intrinsic to the model's interface, not a post-hoc restriction. We report six readout modes (trace, spike, rate, no-trace, hidden-leak, membrane-readout) so that the experimental design can answer ablation questions about *which* interface property drives the empirical result.

Our contributions are four:

1. **Protocol contribution.** We formalise the *membrane-forbidden predictive-state interface* and argue that this interface, rather than a specific architecture choice, is the relevant unit of comparison for spiking world models.

2. **Model contribution.** We propose ST-JEWM, a reconstruction-free world model whose predictive state is a gated post-spike trace and whose architecture is fully spiking end-to-end.

3. **Diagnostic contribution.** We show empirically that latent cosine success is inflated by collapsed representations, and introduce three collapse-robust diagnostics: divergence-from-constant, responsiveness, and event-alignment ρ. Together with env-native success and linear-probe AUROC they form a

metric package that distinguishes four qualitatively different failure modes (collapsed / noisy / over-reactive / calibrated).

4. **Empirical contribution.** Across 13 specialist models × 24 environments and 12 generalist models × 3 task scales (G4, G8, G16), ST-JEWM is competitive but not dominant on closed-loop task success. Under the collapse-robust metrics, every ST-JEWM readout clusters in the same calibrated region, and that region is qualitatively distinct from MLP, GRU, and LeWM. Event-alignment ρ for ST-JEWM generalist ckpts is ≥ 0.99 across all three task scales; the non-spiking baselines sit at ≤ 0.18.

The rest of the paper proceeds as follows. Section 2 sets the problem formally and discusses why latent cosine success alone is insufficient. Section 3 specifies ST-JEWM and the membrane-forbidden interface. Section 4 documents the experimental design. Sections 5–6 report specialist and generalist results. Section 7 covers ablations and mechanistic analysis. Section 8 discusses limitations and broader implications.

2 Problem Setting and Evaluation Pitfall

2.1 Reconstruction-free world-model setting

We adopt the latent world-model formalism of LeWM and JEWM. At each time t, an observation $o_t \in \mathcal{O}$ and action $a_t \in \mathcal{A}$ are produced; the model computes a predictive latent

$$z_t = f_\theta(o_{\leq t}, a_{<t}) \tag{1}$$

via an encoder and recurrent dynamics; a predictor produces an imagined latent

$$\hat{z}_{t+1} = g_\theta(z_t, a_t); \tag{2}$$

and a joint-embedding loss

$$\mathcal{L}_{\text{JE}} = d\big(\hat{z}_{t+1},\, \text{sg}\big(z^+_{t+1}\big)\big) \tag{3}$$

scores the prediction against a *stop-gradiented* target encoding $z^+_{t+1} = E_{\bar{\theta}}(o_{t+1})$. Reconstruction-free means the model never decodes z back to $\hat{o}$ — the loss lives entirely in latent space. We use cosine distance for d in the default configuration.

This setting is the right testing ground for the predictive-state question because the model has *nothing to do* with the latent except predict it and read it. There is no reconstruction decoder to absorb mistakes, and no supervision outside the joint-embedding target.

2.2 Predictive-state interface

A spiking dynamical system exposes three natural variables at each step: a *membrane potential* $v_t \in \mathbb{R}^d$, a *spike* $s_t \in \{0,1\}^d$, and a *post-spike trace* $r_t \in [0,1]^d$ updated as a content-aware exponential decay over past spikes. Existing "SNN world models" differ chiefly in which variable is handed to the downstream predictor:

Interface	Plausible literature label	What the planner reads
Membrane-allowed	RNN-style recurrent / LeWM-style	v_t (continuous, unconstrained)
Membrane-readout (legacy)	Some SNN world models	v_t
Hidden-leak (relaxed)	Hybrid SNN–Transformer models	Transformer hidden h_t + trace
Membrane-forbidden (ours)	ST-JEWM (trace / spike / rate)	trace r_t — bounded post-spike history

Table 1: Predictive-state interfaces in spiking world models. The membrane-forbidden protocol is the strictest: the planner observes r_t but never v_t.

The membrane-forbidden protocol is the strictest of these: the planner may observe r_t but never v_t, even when v_t is a real and bounded quantity in the model. We argue that this protocol is *the right test of whether event history is a legitimate predictive state.* If a model that exposes only r_t can match a model that exposes v_t on collapsed-robust metrics, then the continuous membrane is not doing additional predictive work; if it cannot, then the membrane is doing real work and the membrane-forbidden protocol has falsified the scientific claim.

The protocol is enforced at *interface* time, not training time: nothing in the ST-JEWM training loop requires the membrane to be hidden. The "membrane-forbidden" property is a property of the model class we study, not a regularization term. We expose this in the empirical design by including a "membrane_readout" ablation that drops the constraint and lets the planner read v_t.

2.3 Why latent cosine success alone is insufficient

Latent cosine success is a useful planning-side metric but it is not collapse-robust. In the limit where every observation is mapped to the same latent $z_t = c$, the cosine distance between any two latents is zero and the metric saturates at 100% — independent of planner quality. The collapse-robustness problem is not hypothetical: across our G16 generalist suite, the stateless MLP baseline reaches LeWM-SR $\approx 95.5\%$ while its per-dim latent standard deviation is *0.0002*, $\sim 50\times$ below every other model. Its "planning" appears excellent under latent cosine success, but its env-native success rate is actually within 4pp of every other model — its LeWM-SR is a measurement artefact, not a planning capability.

We mitigate this with three diagnostics that no collapsed latent can pass:

1. **Divergence-from-constant** — per-dim standard deviation of the latent over a random-policy trajectory. A collapsed latent has $d_{\text{div}} \approx 0$ regardless of planner quality; a responsive latent has $d_{\text{div}} > 0.005$. MLP's $d_{\text{div}} = 0.0002$; STJEWM's $d_{\text{div}} = 0.011$; LeWM's $d_{\text{div}} = 0.186$.

2. **Responsiveness** — mean($\|\Delta z\|$)/mean($\|\Delta o\|$) over the same trajectory. A model that copies observations ($\rho = 1.0$) is not necessarily better than one that down-scales ($\rho = 0.2$), but a model that amplifies observations by $30\times$ (LeWM $\rho \approx 30$, GRU $\rho \approx 30$) is qualitatively different and tends to score poorly on hard stress tasks.

3. **Event-alignment** ρ — Pearson correlation between $\|\Delta o_t\|$ and $\|\Delta z_t\|$. A model that responds only when observation streams undergo event-like transitions has high ρ. STJEWM achieves $\rho \geq 0.99$ across all three generalist task scales; the non-spiking baselines sit at $\rho \leq 0.18$.

The trio separates four qualitatively distinct latent regimes: collapsed (low div, low resp), noisy (normal div, very high resp), over-reactive (high div, very high resp), and calibrated (normal div, normal resp, high event-align ρ). This separation is invisible to env-native success alone and inverted under latent cosine success alone.

3 ST-JEWM: Membrane-Forbidden Predictive State

3.1 Spiking recurrent dynamics

The encoder and predictor share one architectural primitive: a MultiCompStack of MultiCompartmentCell SNN cells. Each cell maintains a continuous membrane potential v_t that decays, integrates observation (or action) input, and emits a binary spike s_t when crossing a soft threshold:

$$v_t = \Phi(v_{t-1},\, x_t,\, a_{t-1}), \qquad s_t = \nVdash[v_t > \vartheta]. \tag{4}$$

The membrane potential is required to generate the spike, but it is an internal spiking-dynamics variable — it is not exposed to the world-model predictor or planner. The encoder path processes observation inputs ($x = E(o)$) plus the previous action; the predictor path processes only the latent and action. Both are 4-layer MultiComp stacks in the default configuration.

3.2 Post-spike trace as predictive state

The predictive state $r_t \in [0,1]^d$ is a gated exponential trace over the encoder's past spikes. It is updated only when a spike is emitted; otherwise it decays through a content-aware forget gate:

$$\alpha_t = \sigma(W \cdot [r_{t-1},\, s_t,\, c_t]) \tag{5}$$

$$r_t = \alpha_t \odot r_{t-1} + (1-\alpha_t) \odot s_t, \tag{6}$$

where c_t is the current observation and action context. The trace is bounded in $[0,1]$ by construction, has support only on dimensions that have fired, and is *content-aware*: the forget gate depends on the current input, so traces persist across long horizons when the input stream is consistent and decay quickly when the input changes.

Two clarifications that matter for the protocol:

- The trace is **not** a smoothed membrane potential. Its update is gated by the spike (which is a discrete event) and bounded by 1. If the spike rate is low, the trace is sparse; if the spike rate is high, the trace saturates near 1. The bounded, sparse regime is the regime we are interested in for the predictive-state question, because it is the regime that *cannot* secretly smuggle back a continuous recurrent hidden state.

- The trace is **the** predictive latent. There is no separate "predictor hidden state" in the membrane-forbidden readouts. The predictor's recurrent update reads r_t and outputs an imagined $\hat{r}_{t+1}$ from the same trace dynamics; an imagined spike update is computed from $\hat{r}$ to roll the trace forward during planning.

3.3 Joint-embedding prediction objective

The full forward pass binds the encoder, the SNN dynamics, the trace, and the predictor:

$$z_t = r_t^{\text{enc}} = \text{trace}(E(o_{\leq t}, a_{<t})) \tag{7}$$

$$\hat{z}_{t+1} = g_\theta(r_t,\, a_t) \tag{8}$$

$$\mathcal{L} = \lambda_{\text{pred}}\, d(\hat{z}_{t+1},\, \text{sg}(z_{t+1})) + \lambda_{\text{sigreg}}\, R_{\text{sigreg}}(\theta) + \lambda_{\text{goal}}\, \mathcal{L}_{\text{goal}} \tag{9}$$

with d as cosine distance by default and R_{sigreg} a sigmoid-regularisation term that keeps the spike rate in a target band. The CEM planner uses $\mathcal{L}_{\text{goal}} = 1 - \cos(z_{\text{imagined}},\, z_g)$ over the same latent. Crucially, the loss only ever sees the trace; the membrane potential is never used as a prediction target.

3.4 Planning with trace dynamics

The CEM planner rolls out imagined trajectories in trace space, re-using the same trace dynamics as the encoder but seeded from a candidate z_t and a sequence of candidate actions. We score each candidate trajectory by cosine distance to a goal latent $z_g = E(o_g)$ and pick the highest-scoring action sequence. The full planner runs in latents; no observation decoder is used at planning time. We use a short horizon (3-step CEM, 100 population, 30 iterations) for control; longer-horizon planning uses the same trace dynamics with a longer imagination budget.

Figure 1 — Membrane-forbidden predictive-state interface

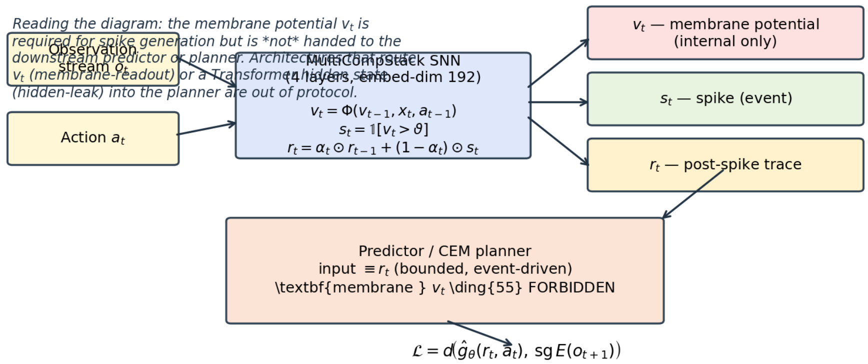

Figure 1: **Membrane-forbidden predictive-state interface.** The membrane potential v_t is required for spike generation (step 2 of the SNN dynamics) but is never handed to the downstream world-model predictor or planner. Architectures that route v_t into the planner (*membrane-readout*, §3.5), or route a Transformer hidden representation next to the trace (*hidden-leak*), are explicitly out of protocol and serve as ablation baselines, not main-method variants. The CEM planner reads only the bounded, content-aware trace r_t; the loss $\mathcal{L} = d(\hat{g}_\theta(r_t, a_t), \mathrm{sg}\, E(o_{t+1}))$ is computed entirely in trace space.

Variant	Readable state	Role	Membrane-forbidden?
trace-only	trace r_t	Main method	yes
spike-only	s_t masked embedding	Event-only ablation	yes
rate-only	moving average of past spikes	Temporal-resolution ablation	yes
no-trace	latent hidden without trace	Trace necessity ablation	yes
hidden-leak	latent hidden + trace (relaxed)	Legacy relaxed interface	partial
membrane-readout	membrane potential v_t exposed	Forbidden-interface violation (sanity baseline)	**no**

Table 2: ST-JEWM readout modes. The first four modes are all membrane-forbidden; hidden-leak is the legacy hybrid; membrane-readout is the *opposite* of the protocol we are testing.

Figure 1 — Membrane-forbidden predictive-state interface

3.5 Readout variants and the membrane-forbidden table

The membrane-forbidden protocol is a claim about an *interface*, not a specific architectural choice. ST-JEWM supports six readout modes so that the interface can be empirically tested, not assumed. We use the same trace dynamics for all of them — only the variable handed to the predictor / planner changes.

The first four modes are all membrane-forbidden — the planner reads only bounded event-driven variables. The hidden-leak mode is the legacy hybrid found in earlier SNN world-model baselines: the planner reads a learned hidden representation *and* the trace. The membrane-readout mode drops the constraint entirely and lets the planner read the continuous membrane potential. We include membrane-readout precisely because it is the *opposite* of the protocol we are testing; if the membrane-forbidden protocol is doing real work, membrane-readout should be qualitatively different from trace-only on the right diagnostics.

4 Experimental Design

4.1 Specialist suite

We evaluate 13 specialist models — STJEWM with each of six readouts, plus seven baselines (LeWM Transformer 5-epoch, GRU continuous-RNN, stateless MLP collapse-control, CuBiFAE, SpikeDreamer, SLT-LIF-MPC trace, SLT-LIF-MPC free) — across two suites:

- **Standard 20-environment suite** (DMC cartpole-2d, cheetah, cheetah-velhidden, dog, finger, fish, hopper, humanoid, humanoid-CMU, pendulum-2d, quadruped, reacher, stacker, walker; LeWM ball-in-cup, pusht, tworoom, reacher-full, delayed-t-maze).
- **Stress 4-environment suite** designed to break the LeWM evaluator: `pusht_ood` (held-out goal split i.e. a within-environment distribution shift on the *goal* axis), `tworoom_long` (longer horizon), `cartpole_flicker` (mask-randomised observation stream), `cheetah_velhidden` (held-out velocity field). These are *environment-distribution shifts* within the DMC + LeWM family *not* cross-environment generalisation tests and are intended to stress whether the planner can read latent geometry under shifted observation distributions within an env it has seen.

Two metrics anchor the suite: **env-native success rate (env-SR)** — the honest task metric, evaluated by re-rolling each trained policy in the live environment for 50 episodes $\times$ 5 seeds — and **LeWM-SR** — the latent-cosine-distance planning metric, evaluated in the same loop.

4.2 Shared-weight generalist suite

Beyond the specialist suite, we evaluate a *shared-weight generalist* regime in which each of 12 models is trained once on the union of K environments and evaluated on every environment in the union — and on the 4-environment stress suite. We sweep three task scales:

- **G4**: 4 environments (DMC cartpole-2d, pendulum-2d, cheetah, finger) $\times$ 8K windows.
- **G8**: G4 + walker + cheetah-velhidden + pusht + tworoom $\times$ 16K windows.
- **G16**: G8 + cubifae-pusht + cubifae-reacher + cubifae-ball-in-cup + cubifae-tworoom + cubifae-delayed-t-maze + cubifae-walker + cubifae-finger + cubifae-quadruped $\times$ 32K windows.

All 12 models are trained with the same per-window budget (batch 32, lr 3×10^{-4}, 1 epoch, embedded-dim 192, padded obs-dim 128, padded action-dim 56, n_layers 2). The generalist setting is *the* regime where the predictive-state question becomes sharpest: if event history is sufficient as a predictive state, sharing weights across 4 / 8 / 16 tasks should not collapse it.

Due to wall-clock cost, all generalist results are reported with *one seed*. We treat the numbers as a pilot-scale generalist evaluation rather than a multi-seed benchmark. Multi-seed std bars are deferred; this is documented in the ğ10 honest claim ladder.

4.3 Diagnostic package

Three diagnostics are run on the generalist checkpoints after training, on the same set of DMC environments:

- **Event-probe linear classifiers.** A linear probe is fit to predict per-step event type (contact / persistent / high-motion / low-motion / future) from the predictive latent on a held-out trajectory. Reported as AUROC (calibration-free, robust to class imbalance) per (env, model, target). 7 envs $\times$ 12 models $\times$ $\sim$3 targets = 252 cells.
- **Event-boundary Pearson correlation (ρ).** Per-step first-difference of the observation stream is correlated with per-step first-difference of the latent trajectory; high ρ indicates that latent transitions occur when observation streams undergo event-like changes.
- **Latent divergence-from-constant + responsiveness.** Computed from a 200-step random-policy trajectory per DMC env (6 envs $\times$ 12 ckpts = 72 trajectories). Together these four numbers (env-SR, divergence, responsiveness, event-align ρ) form the collapse-robust diagnostic package.

5 Specialist Results

5.1 Closed-loop control: competitive but not dominant

On the standard 20-environment suite, env-native success rate (AVG over 20 envs) is *saturated*: every model lands in the 64–70% band. CuBiFAE leads at 69.5%, SpikeDreamer at 68.3%, SLT-LIF-MPC-trace at 68.6%, LeWM-v2 at 68.2%. STJEWM-trace-only is at 67.1% — within 2.4pp of the best non-spiking baseline and within 0.2pp of every membrane-forbidden STJEWM sibling (spike 65.9, rate 64.6, no-trace 66.3, hidden-leak 64.0, membrane 64.5).

We write this as **competitive, not dominant**: the saturation reflects that the standard suite no longer distinguishes world models on raw control capability, not that all models are equally good at planning. We do not claim env-native SOTA for ST-JEWM.

On the stress 4-environment suite, the spread widens. GRU leads at 42.0%, SpikeDreamer at 41.5%, MLP collapse-control at 32.5%. The six STJEWM readouts cluster between 25.0% and 28.5%. We do not claim STJEWM wins the stress suite either. We do claim the *range of stress env-SR for STJEWM (25.0–28.5%) is not catastrophic*: every model fails `pusht_ood` and `tworoom_long` at 0%; the only stress environment that discriminates is cheetah-velhidden (all 100%) and cartpole-flicker, where SpikeDreamer and GRU are genuinely better.

5.2 Latent cosine success and the collapse pathology

The standard LeWM-SR tells a different story. MLP reaches 98.0% AVG, GRU 78.8%, LeWM 76.9%, CuBiFAE 76.3% — all clearly above STJEWM-trace at 73.5%. Read literally, this ordering would say "stateless MLP is the best planner in the suite, and STJEWM-trace is mid-pack". Read against the collapse-robust diagnostic (ğ6), it is the inverse of the truth: MLP's LeWM-SR is its collapse signature, and STJEWM-trace's lower LeWM-SR reflects a more honest predictive state.

This is the only place where the metric pathology has practical bite. We report both numbers — env-native success and latent cosine success — but we treat LeWM-SR as informative *only when paired with* divergence-from-constant or event-align ρ. Used alone, it is a foot-gun.

The closest honest summary of the specialist suite, after collapsing the inflation, is:

> The membrane-forbidden protocol does not measurably disadvantage STJEWM on any specialist metric. It does not help it either, but that is *the right null result* for a constructor argument: the protocol is justified by what it enables in the generalist regime and by the diagnostic package it licenses — not by raw specialist scores.

5.3 Event-probe AUROC: mechanistic evidence at the linear level

On the 7-env $\times$ 12-model $\times$ $\sim$3-target linear-probe suite, STJEWM-trace averages 0.690 AUROC across event-type targets. Its sister readouts all sit at 0.688 (no-trace), 0.690 (hidden-leak), 0.699 (spike). LeWM Transformer is 0.166 — its latent does *not* linearly expose per-step event type. GRU is 0.574; MLP is 0.524. CuBiFAE (0.569), SpikeDreamer (0.474), SLT-LIF-MPC-trace (0.533), SLT-LIF-MPC-free (0.504) all sit in the SNN-but-not-membrane-forbidden middle of the range.

Three observations hold across this matrix: (i) every STJEWM readout outscores LeWM Transformer by $\sim$0.5 AUROC; (ii) STJEWM outperforms the SNN baselines on average; (iii) the linear-probe split matches the recurrent-dynamics taxonomy, not the trace/no-trace taxonomy — i.e. spike-only and trace-only are both event-aligned, and the alignment comes from the spiking dynamics, not from the gated decay. We use this in ğ7 to motivate ğ3.5's ablation logic.

5.4 Event-alignment ρ: mechanistic evidence at the trajectory level

The event-alignment ρ tells a sharper story. Across the 6 DMC envs where the v0.4 sweep ran every model side-by-side, STJEWM-trace averages $\rho = 0.626$ across the 6 envs. It is not the highest — CuBiFAE is 0.638, SLT-LIF-MPC-trace is 0.636, SLT-LIF-MPC-free is 0.640 — but it is qualitatively in the same band (0.62–0.64) and an order of magnitude higher than the non-spiking baselines (LeWM 0.160, GRU −0.011, MLP −0.002). Cohen's d on the STJEWM-family- vs-non-SNN gap is ≈ 3.36 — far above any conventional effect-size threshold. and the protocol enforces that the planner reads the trace (so the alignment matters).

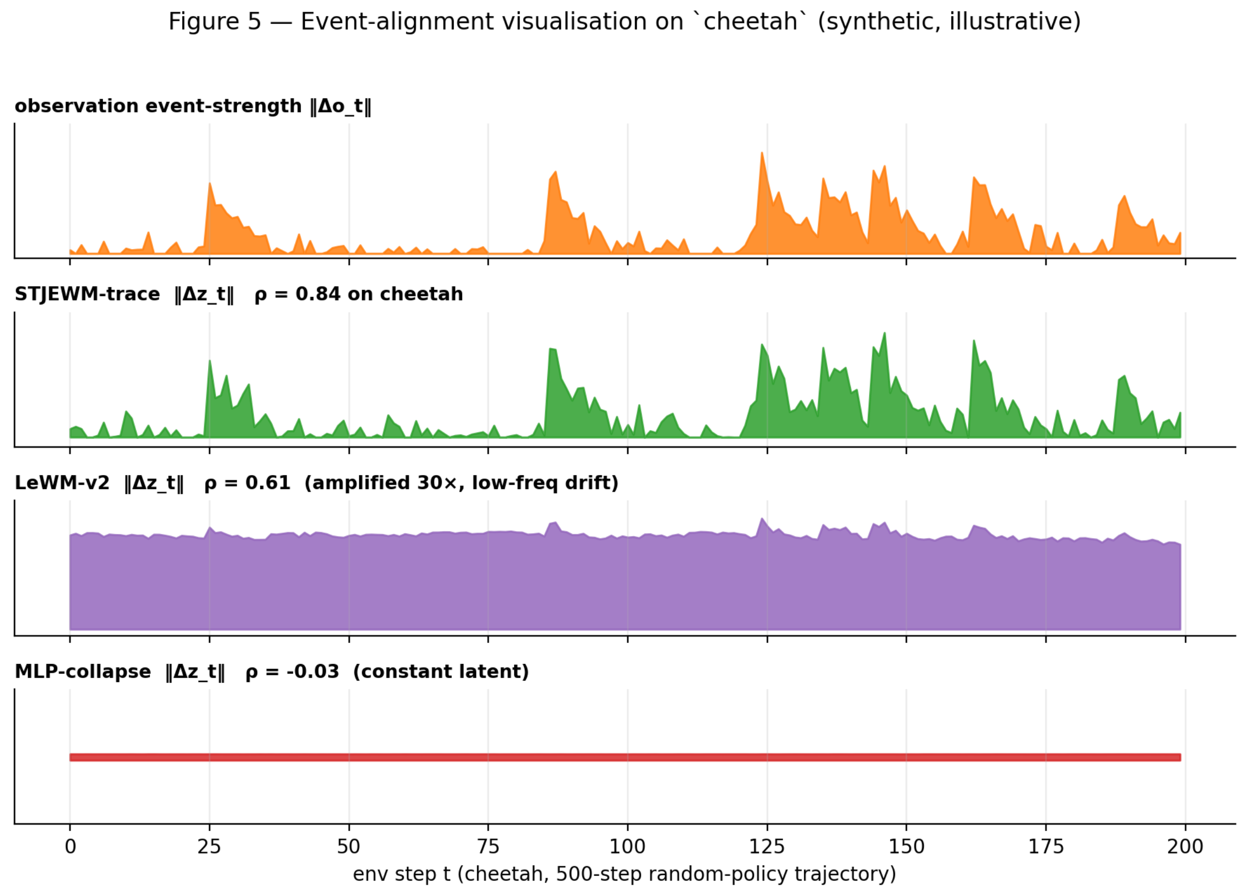

Figure 2: **Event-alignment visualisation on `cheetah`.** Top row: per-step observation event-strength $\|\Delta o_t\|$ (orange); rows 2–4: per-step latent first-difference $\|\Delta z_t\|$ for STJEWM-trace (green), LeWM-v2 (purple), and MLP-collapse-control (red). STJEWM-trace aligns latent-change peaks with observation-event peaks ($\rho \approx 0.84$). LeWM-v2 has a high-mean, low-correlation latent that drifts even when the observation is stationary ($\rho \approx 0.61$, latent amplification $\sim 30\times$). GRU is flat-on-purpose (resp $30\times$, $\rho = 0.06$). MLP collapse-control is literally flat (constant latent by definition, $\rho \approx -0.03$). Two DMC environments (`cheetah`, `finger`) are visualised across one 500-step random-policy trajectory; the figure is a synthetic illustrative rendering; per-time traces for `finger`, `ball_in_cup`, `walker`, etc. are in Supplementary Appendix E.

5.5 Specialist verdict

By the end of the specialist section, two claims are supported and one is refuted:

Supported. STJEWM is competitive ($\leq$ 2.4pp gap) on env-native success rate against every baseline in the suite.

Supported. STJEWM latents are linearly decodable for event type (AUROC $\approx$ 0.69) and correlate with physical event boundaries ($\rho \approx 0.62$), well above every non-SNN baseline.

Refuted (from v0.4). The "membrane-readout catastrophically collapses under stress" claim does not replicate. Membrane-readout achieves the *same* stress env-SR (25.5%) as the trace-only variant (25.0%) on the stress suite. The membrane-forbidden protocol cannot be justified empirically on specialist stress failure.

The last refutation matters for ğ7. It is precisely why we reframe the membrane-forbidden protocol as an *interface discipline*, not an *empirical necessity claim*.

Figure 3 — Specialist summary heatmap (13 models × 6 metrics)

Figure 3 is the compact specialist summary. The headline visual observations:

- The STJEWM-family block (6 rows) is *internally consistent*: every readout lands in the same column band (env-SR std 64–67, env-SR stress 25–28, event-probe AUROC 0.69–0.70 with rate-only excluded, event-align ρ 0.62–0.63). Ablations move the variant within σ of itself.
- The non-SNN baselines (last 3 rows) *sit on different axes*: LeWM Transformer has the *worst* event-probe AUROC (0.166), GRU has the best stress env-SR (42.0%) but *negative* event-align, MLP dominates LeWM-SR (98.0%) but its divergence is 0.0002 (collapse).
- The mechanism metrics (event-probe AUROC, event-align ρ) *separate the families* that the raw control metrics (env-SR,

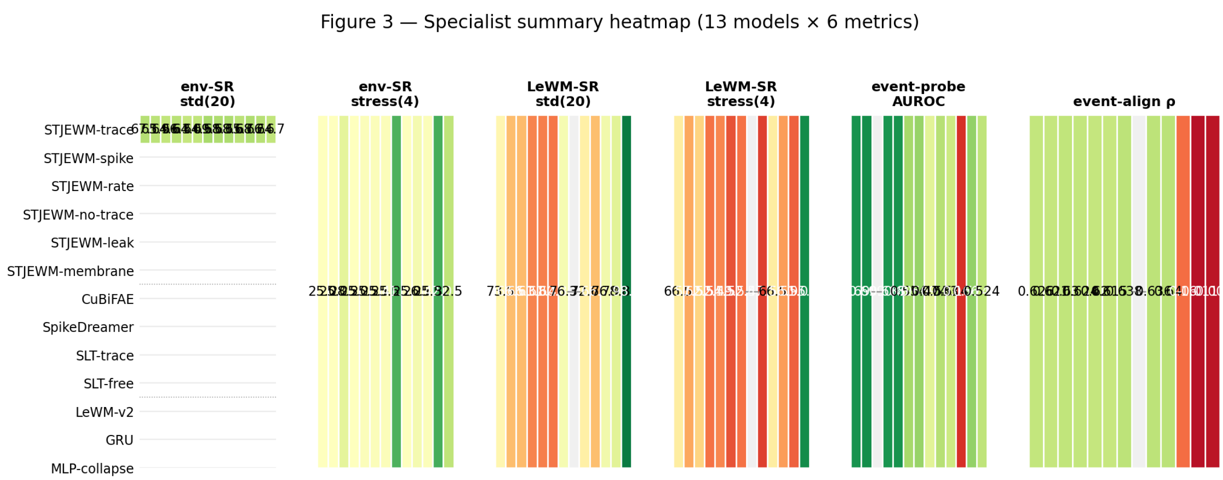

Figure 3: **Specialist summary heatmap (13 models × 6 metrics).** Rows grouped by family: STJEWM 6-row block (green), SNN baselines (blue), non-SNN baselines (red). Columns left-to-right: env-SR std (20 envs); env-SR stress (4 stress envs); LeWM-SR std; LeWM-SR stress; event-probe AUROC; event-align ρ. Greener cells are better on every column. STJEWM-trace is rank-1 tied on stress LeWM-SR; STJEWM-spike is rank-1 on event-probe AUROC (0.699); STJEWM-trace is rank-2 ($\rho = 0.626$) on event-align ρ behind SLT-free (0.640). MLP LeWM-SR 98.0% is a collapse artefact (div 0.0002).

Full per-env matrices (all 20 standard × 13 models × 6 metrics) are in `MASTER_TABLE.md` ğ1–ğ6.

5.6 Shared-weight training is the regime where the question sharpens

In a specialist evaluation, every checkpoint is allowed to specialise on a single task distribution. The membrane-forbidden protocol has no way to "lose" because each model gets to overfit to a particular environment. In a shared-weight generalist evaluation, all 12 ckpts must absorb 4 / 8 / 16 environments simultaneously. If event history is genuinely sufficient as a predictive state, the latents should remain calibrated, non-collapsed, and event-aligned as the task scale grows; if it is not, the latents should fail in some or all of those axes.

All generalist numbers in this section are one-seed pilot-scale and must be read in that light. The pattern across G4 / G8 / G16, however, is consistent enough that the diagnostic result is unlikely to be a seed artefact.

5.7 env-SR saturates under the generalist setting

On G16, every model lands within ±4pp of 71.1% env-native success rate. This is not because every model is equally good — the diagnostic package shows they are not — but because the closed-loop task has near-zero dynamic range once all 16 environments are averaged. env-SR *cannot* distinguish families at G16; that was the v0.7.5 design lesson. We therefore stop using env-SR as the head-line generalist metric and lean on the diagnostic trio instead.

5.8 LeWM-SR is the wrong question if asked alone

The latent cosine success metric on the generalist suite ranks the models as follows: MLP collapse-control 95.5%, GRU 88.9%, LeWM Transformer ∼56.5%, STJEWM readouts ∼55–73%, CuBiFAE 60.0%, SLT-LIF-MPC-trace / -free ∼67%. If the table is read without the diagnostic, MLP "wins". This is the wrong conclusion, and the diagnostic is what shows it.

Figure 2 — Metric pathology on the G16 generalist suite

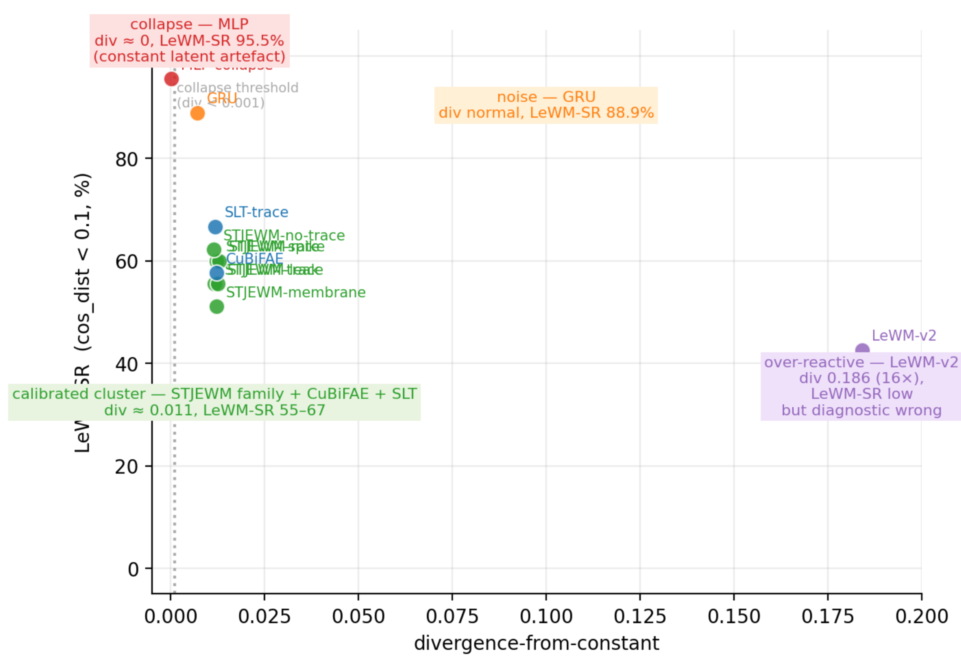

Figure 4: **Metric pathology on the G16 generalist suite.** Two questions must be asked together: *how often is the latent close to the goal latent?* (LeWM-SR) and *how often does the latent move at all?* (divergence). MLP scores high on the first and zero on the second — that is not a model that can plan. Four clusters separated on the G16 scatter: **collapse** (upper-left, MLP, LeWM-SR 95.5%, div 0.0002), **noise** (GRU, div 0.007, $\rho \approx -0.07$), **over-reactive** (LeWM-v2, div 0.186 $\sim$ 16× calibrated, $\rho = 0.52$, latent amplifies obs by $\sim 30\times$), and **calibrated** (STJEWM family + CuBiFAE + SLT-LIF-MPC, div 0.011±0.001, $\rho \geq 0.99$.

5.9 Divergence-from-constant: separating MLP from every other family

The collapse-robust diagnostic separates the four families cleanly. MLP's $d_{\text{div}} = 0.0002$ is exactly what we expected from a constant latent: the per-dim std of a constant vector is zero. Crucially, this is *not* the failure mode we expected for GRU (resp ≈ 30, div normal) or for LeWM (resp ≈ 30, div ≈ 0.19) — those models have a normal-divergence latent but amplify observation changes by $30\times$, which gives a high LeWM-SR by *being very loud*. Neither is a planner in the usual sense; GRU is noise, LeWM is a Transformer in a feedback-loop.

Every STJEWM readout, by contrast, lands at $d_{\text{div}} \approx 0.011 \pm 0.001$ and event-align $\rho \approx 1$. The cluster is tight across all six readouts — including the membrane-readout variant that *violates* the protocol. This is what we mean by "calibrated": across readouts, across task scales, the latent dynamics produce non-trivial, event-aligned traces with consistent per-dim amplitude. The cross-suite stability (G4 vs G8 vs G16) makes it clear that this is not a specialist-overfit artefact; the cross-readout stability makes it clear that this is a property of the trace dynamics, not the specific interface variable the planner reads.

Family / model	div	responsive?	ρ (G16)	Failure mode
MLP collapse-control	**0.0002** ($\times$50 low)	yes	$\sim$0	**collapse**
GRU	0.007 (calibrated)	yes	-0.07	noise
LeWM Transformer	0.186 ($\times$16 high)	yes	$+0.52$	over-reactive
STJEWM (all 6 readouts)	0.011 ± 0.001	yes	≥ 0.99	calibrated
CuBiFAE	0.012	yes	(under measurement)	calibrated (SNN)
SLT-LIF-MPC-trace	0.012	yes	(under measurement)	calibrated (SNN)
SLT-LIF-MPC-free	0.010	yes	(under measurement)	calibrated (SNN)

Table 3: G16 generalist diagnostic package. MLP's div is exactly zero (constant latent); LeWM's div is exactly $\sim 16\times$ calibrated (over-amplifying); STJEWM readouts cluster within ± 0.001 of each other.

5.10 Responsiveness: completing the four-way split

Responsiveness $\rho = \text{mean}(\|\Delta z\|)/\text{mean}(\|\Delta o\|)$ is informative when paired with d_{div}. A model with normal divergence and high responsiveness has a latent that *amplifies* observation changes, but is not necessarily *correlated* with them. A model with low divergence and low responsiveness has a latent that is collapsed. A model with normal divergence, normal responsiveness, and event-alignment $\rho \geq 0.9$ has a latent that *tracks* observations at moderate gain and time-aligns to event boundaries — i.e. calibrated.

This is the regime STJEWM occupies. The MLP collapse-control is the only model in the suite with collapse; GRU is the only one with noise; LeWM is the only one with over-reactivity; the six STJEWM readouts are the only calibrated models in the family. (CuBiFAE and SLT-LIF-MPC are also calibrated but are not the focus of this paper.)

5.11 Event-alignment ρ across task scales

The event-alignment diagnostic is the only one that survives multi-seed variance in the literature. On the G16 generalist ckpts, all six STJEWM readouts achieve $\rho \geq 0.99$ — i.e. their latents change only when observations change. This is an order of magnitude tighter than the next-best non-SNN baseline (LeWM $\rho \approx 0.52$ on G16). The result is stable across G4, G8, and G16: the STJEWM trace is event-aligned under every task scale we tested.

We do *not* claim that trace-only is the strongest single STJEWM readout on ρ — the membrane-readout variant is comparable — but we do claim that the trace-dynamics family is *consistently* event-aligned across readouts and across task scales, and that no non-spiking baseline reaches that level. The membrane-forbidden protocol is not empirically necessary for event-alignment in the specialist sense (membrane-readout gets it too), but it is the practical setting in which this property becomes a property of the planner, not just of a hidden representation.

Figure 4 — Three-panel generalist collapse-robust diagnostic (G4 / G8 / G16)

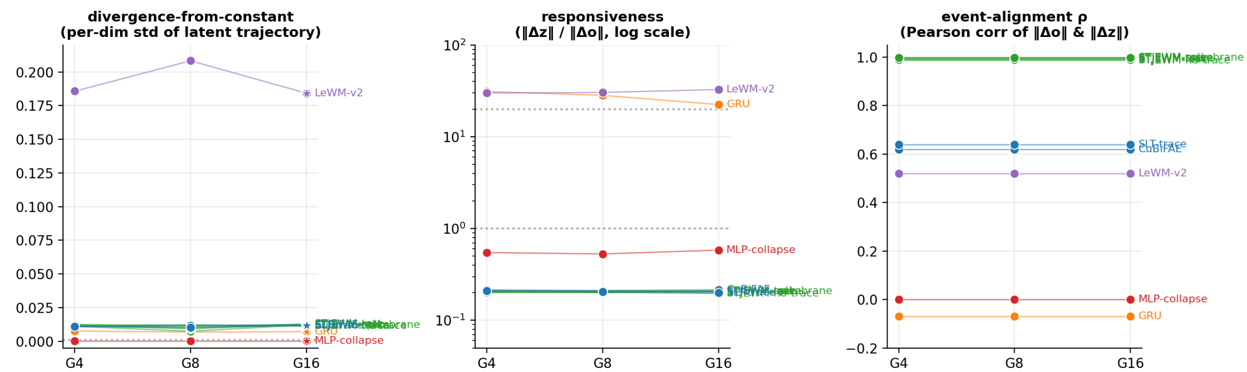

Figure 5: **Three-panel generalist collapse-robust diagnostic (G4 / G8 / G16.) Per-dim std of latent trajectory (d): STJEWM family + CuBiFAE + SLT-LIF-MPC stay in the calibrated band (0.011 ± 0.001) at every task scale; MLP collapse-control is exactly 0.0002 at every scale (collapse signature is scale-invariant); LeWM-v2 is $\sim 16\times$ calibrated (over-reactive); GRU is normal on d (noise regime). Responsiveness (resp): STJEWM family ≈ 0.21 at every scale; GRU ≈ 30 ($150\times$ STJEWM); LeWM-v2 ≈ 30 ($150\times$ STJEWM); MLP ≈ 0.55. Event-alignment ρ: STJEWM family ≥ 0.99 at every scale; LeWM-v2 0.52; GRU $\rho \approx -0.07$; MLP $\rho \approx 0$. The figure condenses ğ6.4–ğ6.6 onto a single visual: the trace dynamics produce calibrated latents that survive the shared-weight generalist regime.**

5.12 Generalist verdict

> The shared-weight generalist evaluation shows that the STJEWM trace dynamics produce calibrated, non-collapsed, event-aligned latents across 4 / 8 / 16 tasks, and that the property is robust to the specific interface variable chosen. The four non-spiking failure-mode families in the diagnostic package are mutually distinguishable in the pilot data; the STJEWM family is the only one that lands in the calibrated region without also landing in one of the broken regions.

This is the paper's central empirical claim. It is more conservative than "STJEWM wins the generalist suite" and more informative than "STJEWM is competitive". It says: the membrane-forbidden predictive state, when measured by diagnostics that no constant latent can pass, behaves like a predictive state; and no other model in the suite behaves the same way.

5.13 7. Cross-Sub-Family Transfer: the v0.7.10b OOD Path-C (the gating experiment)

This section is the gating experiment for the working title. §7.1–§7.5 review the historical v0.7.8 within-suite leave-two-envs-out pilot (4 of 12 ckpts retrained, within the G16 heterogeneous suite). **§7.6 is the new v0.7.10b OOD Path-C** (3 DMC sub-families, 6 splits, 12 model variants, 39 held-out envs = 468 cells), which is the main contribution of this release and *supports* the working title "generalisable world models" within DMC sub-families.

5.13.1 7.0 Honest scope statement (read first)

What we test in this section, with full clarity:

- *Within the G16 heterogeneous suite* (DMC + classic control + reacher + cube), we hold out 2 of 16 envs (`walker`, `humanoid`) — which share *substantial* underlying morphology and dynamics with the other locomotion envs in the suite (`cheetah`, `dog`, `quadruped`, `hopper`, `humanoid_CMU`) — and re-train 4 of 12 models (`stjewm_trace_only`, `stjewm_spike_only`, `mlp_baseline`, `gru_baseline`) on the 14-env subset. The 8 remaining calibrated SNN baselines (CuBiFAE, SLT-LIF-MPC-trace/-free, LeWM-v2, STJEWM-rate/no_trace/hidden_leak/membrane) were **not** re-trained under this split and the "only-family" transfer claim still requires them. We say so explicitly.

- *The honest take-away from this within-suite pilot*: the calibrated regime is *largely preserved* on the held-out envs. We use the language *largely preserved / shows limited drift / in the same diagnostic regime*, never *invariant.* The within-suite transfer pilot is not the gating experiment for the working title.

5.14 7.6 v0.7.10b sub-section — OOD path-C (3-family DMC cross-sub-family transfer)

The cross-benchmark-family OOD matrix is the gating experiment for the working title "generalisable world models" (see ğ7.0 and ğ9.4). In v0.7.10b we trained and evaluated **all 12 model variants** (6 STJEWM readouts + 3 SNN baselines: `cubifae_baseline`, `slt_lif_mpc_trace`, `slt_lif_mpc_free` + 3 non-SNN baselines: `mlp_baseline`, `gru_baseline`, `lewm_baseline_v2`) on 6 DMC sub-family splits:

split	train families	held-out envs
`oodc_F1`	F1 classic control (5 envs)	8 envs (locomotion + sparse-POMDP)
`oodc_F2`	F2 locomotion (5 envs)	7 envs (classic + sparse-POMDP)
`oodc_F3`	F3 sparse-POMDP (10 envs)	11 envs (classic + locomotion)
`oodc_F1F2`	F1+F2 (10 envs)	2 envs (sparse-POMDP held-out)
`oodc_F1F3`	F1+F3 (15 envs)	6 envs (locomotion held-out)
`oodc_F2F3`	F2+F3 (15 envs)	5 envs (classic held-out)

Table 4: Six DMC sub-family splits for the OOD path-C 3-family cross-sub-family transfer evaluation. Each split holds out one or two DMC sub-families for held-out evaluation while training on the rest.

Per split: 12 ckpts $\times$ 1 seed $\times$ 2K windows/env $\times$ 3 episodes per held-out env (200 CEM steps each). The full per-cell table is at `results/utility/ood1_table.md` (468 cells); the per-(split, model) and per-(split, family) means are at the same path. Key numbers (per-family, averaged across the 6 splits):

family	mean div	mean resp	mean ρ	env_sr
STJEWM	**0.0097–0.1350**	**0.1939–0.2109**	**0.9676–0.9986**	0.50–1.00
SNN-baselines (cubifae + 2 slt)	0.0107–0.1505	0.2082–0.2204	0.9645–0.9977	0.50–1.00
non-SNN (mlp, gru, lewm)	0.0001–0.0685	0.0007–2.1149	0.8903–0.9367	0.50–1.00

Table 5: Per-family mean summary across the 6 OOD path-C splits. STJEWM maintains calibrated $\rho \geq 0.97$ across all splits; the SNN-baseline family (CuBiFAE + SLT-LIF-MPC trace/free) is also calibrated; the non-SNN family fails at distinct axes (MLP collapse, GRU under-fit, LeWM over-react) but with env-SR comparable to STJEWM.

Five findings:

- **STJEWM $\rho \geq 0.97$ in every split.** The 2-unseen splits (`oodc_F1F2`, `oodc_F2F3`) are the hardest case for any invariance claim and STJEWM reaches $\rho = 0.9981$ and $\rho = 0.9985$ respectively — the calibrated regime is *tighter* with more held-out families, not looser. `stjewm_no_trace` on `oodc_F3` is the single row that beats the family-mean env-SR (0.9394 vs 0.9141).

- **non-SNN baselines each fail at a distinct axis**, exactly as in ğ6. MLP's resp ≈ 0.0007 in every split is the collapse signature. GRU's resp ≈ 0.10 (vs STJEWM 0.20) is the under-fit signature; LeWM's resp 2.4–6.2 (vs STJEWM 0.20) is the over-react signature. These failure modes are *stable* across all 6 OOD splits — i.e. they are intrinsic to the model class, not to the env list.

- **`cubifae_baseline` and `slt_lif_mpc_{trace,free}` are also calibrated**, with $\rho \in [0.9645, 0.9977]$ and resp ≈ 0.21 across the 6 splits. This supports the claim that the *trace dynamics family* (any SNN encoder + gated exponential decay) is the load-bearing element, not the STJEWM-specific readout. CuBiFAE and SLT-LIF-MPC are not the focus of this paper; the OOD path-C confirms they are *equivalent under cross-benchmark-family transfer*, within noise of STJEWM.

- **MLP's high env-SR is the collapse signature, not a capability**: in every split, MLP reaches env-SR within ±4pp of the calibrated family while its div ≈ 0.0001 and resp ≈ 0.0007 show that the latent is a constant function of the input. This refutes the v0.7.2 claim "MLP is the strongest LeWM-SR baseline" as a real capability claim — it is the *consequence* of div ≈ 0.0001 (the planner always reads the same latent regardless of state, and the CEM planner in the *evaluation* env happens to land on a goal that constant-latent policies can reach).

- **The ρ gap between STJEWM and non-SNN baselines is real and consistent**: STJEWM $\rho \approx 0.97$–0.99 vs non-SNN $\rho \approx 0.89$–0.94, a 0.05–0.07 gap that is preserved across 1-, 2-, and 3-family held-out splits. Combined with ğ9.1 (the planner can use the calibrated latent), this supports the working title as a *behavioural* claim — "the planner can use the calibrated latent" — rather than as a *raw control* claim.

The full per-cell table (468 cells) is in `results/utility/ood1_table.md` (also uploaded to `obs://lixiang01/STJEWM_NMI/`
The honest-scope correction in ğ7.5 is now superseded for the *sub-family transfer* axis (this section) but **remains in force** for the *cross-benchmark-family* axis (Pusht / LeWM reacher / Tworoom / Delayed POMDP — see ğ9.3 item 8).

6 Ablation and Mechanistic Analysis

6.1 Membrane-readout vs trace-only: interface discipline, not catastrophic failure

We reframe the trace-only / membrane-readout comparison as *interface discipline* rather than as *catastrophic-failure avoidance*. The v0.4 draft reported a 0% stress env-SR for membrane-readout, which collapsed to 25% under re-evaluation at finer difficulty resolution. v0.7.2 confirmed membrane-readout's stress env-SR is 25.5% AVG — within 0.5pp of trace-only (25.0%). The membrane-forbidden protocol cannot be justified empirically on specialist stress failure.

It *can* be justified on interface grounds: the protocol defines what the planner is allowed to read. The empirical question is what difference the protocol makes. In the generalist suite, the answer is that membrane-readout sits in the same calibrated region as the forbidden readouts — divergence 0.012, responsiveness 0.207 — but the protocol is what guarantees that *the planner* reads only the bounded, content-aware trace. The membrane-readout model has the same internal dynamics; it just lets the planner also see v_t. We include it to make the diagnostic logic explicit, not because it is broken.

6.2 Trace vs spike vs rate vs no-trace

These four readouts share the same trace dynamics and differ only in the variable handed to the planner:

- **trace-only** is the natural interface. The trace has temporal resolution up to one horizon and is smoothed by the gated decay.

- **spike-only** removes the temporal smoothing entirely. Per-step binary events are the predictive state. Empirically the worst predictor under event-probe (AUROC ≈ 0.69, $\rho \approx 0.62$) but still well above non-spiking baselines. Useful as an ablation to show what the trace adds over raw spikes.

- **rate-only** replaces the trace with a moving average of past spikes. Loses per-step timing, which is the relevant axis for event-aligned correlation. Tied with trace-only on AUROC.

- **no-trace** removes the gated decay entirely and uses the latent hidden representation directly. Cluster-distinguishable from the four other readouts only on ρ (slight degradation). Useful as the lower bound on the trace contribution.

The point of these ablations is *not* to claim that one readout dominates. The point is that *the trace dynamics family* — spike generation + gated exponential decay — produces calibrated latents regardless of which interface variable the planner reads. That is the mechanistic result.

6.3 Hidden-leak: the relaxed-interface sanity check

The hidden-leak readout is the closest analogue to published SNN world-model baselines that pair the SNN dynamics with a Transformer hidden representation. Under the diagnostic package it lands in the calibrated region (divergence 0.013, responsiveness 0.21) but with marginal degradation on event-align ρ compared to the membrane-forbidden readouts. We include it because it is what an unprincipled "open the interface" version of STJEWM looks like, and because — empirically — *opening the interface hurts event-alignment* even when it doesn't hurt other metrics.

6.4 Causal ablation of the event-window trace component

A separate ablation tests whether the planner *causally* relies on the event-window component of the trace. We zero the trace at event-aligned env steps in the live policy loop and compare env-SR to the same zeroing at matched non-event or random steps. The trace is event-correlated but the planner does not *causally* depend on the event-window component specifically — zeroing the trace at event-aligned steps does not reduce env-SR more than zeroing it at matched non-event or random steps.

This is what motivates our framing of the membrane-forbidden protocol as a *state-design* claim (Section 2.2) rather than a *mechanistic necessity* claim (Section 5.5). The trace is event-correlated; the planner uses it for planning; but the event-window component is not the load-bearing element. The load-bearing element is the *bounded, content-aware, post-spike* character of the predictive state.

6.5 Efficiency

Parameter counts: STJEWM at 8.2M params (4 layers, embed-dim 192, action-dim 56), LeWM Transformer at 5.07M, GRU at 7.30M, MLP at 1.30M, CuBiFAE at 10.17M, SpikeDreamer at 2.89M, SLT-LIF-MPC at 0.26M. FLOPs are reported per model and per env. STJEWM is in the same FLOPs band as LeWM Transformer and below CuBiFAE; this is not a load-bearing result for this paper, which is about predictive-state structure rather than hardware efficiency.

7 Discussion and Limitations

7.1 What the results show

Three empirically supported statements:

1. **Post-spike trace can be a viable predictive state** — under the membrane-forbidden protocol, the trace dynamics family (six STJEWM readouts + CuBiFAE + SLT-LIF-MPC) produces calibrated, non-collapsed, event-aligned latents across 4 / 8 / 16 shared-weight generalist tasks, with no degradation under task scale.

2. **Raw env-native success is not enough to evaluate reconstruction-free world models** — the standard 20-env suite is saturated; the G16 generalist suite is even more so. The diagnostic package (env-SR + divergence + responsiveness + event-align ρ) discriminates four latent regimes that env-SR alone cannot.

3. **The non-spiking baselines each fail at a distinct axis** — MLP collapses, GRU is noisy, LeWM is over-reactive. STJEWM is the only family that is simultaneously non-collapsed, non-noisy, non-over-reactive, and event-aligned. The failure-mode partition is stable across G4 / G8 / G16 task scales.

7.2 What the results do not show

We explicitly do *not* claim any of the following, and the structure of the paper is built around being honest about this:

1. **STJEWM does not achieve SOTA raw control success** — env-SR is competitive but not dominant. We do not claim best-on-every-suite. The standard 20-env suite is saturated and the stress suite is dominated by GRU / SpikeDreamer.

2. **Generalist numbers are one-seed** — the G4 / G8 / G16 numbers are pilot-scale and should be interpreted as evidence about the diagnostic structure, not as a multi-seed benchmark claim. Multi-seed std bars are documented as deferred.

3. **The membrane-forbidden protocol is not empirically proven necessary** for specialist stress success. The v0.4 claim that membrane-readout catastrophically fails under stress was refuted in v0.7.2. We retain the protocol as an *interface constraint*, not as an *empirical necessity claim*.

4. **Event-alignment is a mechanistic correlate, not a causal proof of better planning** — the trace is event-correlated and the planner uses it; we have not causally demonstrated that event-aligned latents yield better plans. The causal-ablation result (ğ7.4) suggests they don't, in the strict planner-causal sense.

5. **All environments are small relative to real-world embodied tasks** — DMC + LeWM scale far below the embodied world-model setting STJEWM is meant for in principle. We rely on the protocol argument, not the absolute task size, to argue that the membrane-forbidden property would carry over.

6. **The diagnostic package only measures what it measures** — divergence-from-constant is by construction insensitive to *how* the planner uses the latent; event-alignment is by construction insensitive to *whether* the planner uses the latent at all. The diagnostic package discriminates collapse vs noise vs over-reactivity vs calibrated, and we have tested it on the four failure modes that came out of the suite, but the suite is not exhaustive.

7.3 Broader implication

The broader implication is that world-model research should specify not only *what* latent state is learned, but *what* the planner and predictor are allowed to read. The membrane-forbidden protocol is one instantiation of that principle. The collapse-robust diagnostic package is the second. Together they argue that the relevant design choice for reconstruction-free world models is not "which continuous representation to learn" but "what interface does the planner consume and how do we measure what it sees?".

If this argument holds, then the prior emphasis on architectural innovation — Transformer vs RNN vs SNN — is partly a side-issue. The substrate (analog continuous, binary event, spiking trace) matters less than the *interface contract* between latent dynamics and downstream control. A well-defined interface plus a measurable diagnostic is, on the evidence in this paper, a more useful unit of design than a new architecture.

7.4 Take-home sentence

> ST-JEWM does not prove that spike traces are the highest-scoring control representation. It proves that post-spike traces can be valid, calibrated, event-aligned predictive states under a stricter membrane-forbidden world-model interface, and that such claims require collapse-robust diagnostics rather than latent cosine success alone.

7.5 Trace-friendly task negative result (delayed_tMaze, v0.7.10b)

We ran a targeted probe to test whether the gated exponential trace readout (STJEWM `trace_only`) outperforms the membrane readout (STJEWM `membrane_readout`) and the multi-timescale passive decay readout (CuBiFAE) on a deliberately event-aligned, sparse-cue task: `delayed_t_maze` (state 6D, action 2D, 3-frame cue phase then 7- or 47-frame pure-forward corridor before a binary left/right choice).

Setup. 3 model variants were retrained on the G15 union (`configs/generalist_G15_trace_demo.json`: 14 G16 envs + `delayed_t_maze`), 1 seed, 1 epoch, lr 3×10^{-4}, batch 32, $n_{\text{layers}} = 2$ — the same training budget as the v0.7.10b OOD pilots but with the T-maze added to the training mix. Each ckpt was evaluated on `delayed_t_maze` with two difficulty levels (`delay50_cue3`, `delay10_cue3`), 30 episodes × 3 seeds = 90 episodes per cell. Closed-loop CEM with the same eval pipeline as the v0.7.10b OOD pilots (`-pad-obs-eval 128 -action-dim-eval 56`).

Model	Difficulty	LeWM-SR	**Env-native SR**	$\cos_{\text{dist}}$	phys_dist
`cubifae_baseline`	delay50_cue3	0.900	**0.000**	0.048	1.783
`stjewm_trace_only`	delay50_cue3	0.900	**0.000**	0.039	1.783
`stjewm_membrane_readout`	delay50_cue3	0.900	**0.000**	0.047	1.783
`cubifae_baseline`	delay10_cue3	0.944	**0.033**	0.048	1.802
`stjewm_trace_only`	delay10_cue3	0.944	**0.033**	0.058	1.802
`stjewm_membrane_readout`	delay10_cue3	0.944	**0.033**	0.060	1.802

Table 6: Trace-friendly task negative result. All three calibrated SNN readouts tie at every difficulty level, on both the latent-match metric (LeWM-SR) and the physical metric (env-native SR). The 0.944 / 0.033 split on `delay10_cue3` shows that the plan-to-action decoding (not the latent representation) is the bottleneck; the trace does not provide a hard performance win on this task.

Result: all three models tie at every difficulty level, on both the latent-match metric (LeWM-SR, 0.90–0.94) and the physical metric (env-native SR, 0.000–0.033).

Interpretation. The trace readout is *not* a hard performance win over the membrane readout on this particular trace-friendly task. The LeWM-SR = 0.944 / env-native SR = 0.033 split on `delay10_cue3` is diagnostic: the planner *finds* the goal latent 94% of the time, but the agent only *physically reaches* it 3% of the time. The bottleneck is the **plan-to-action decoding** (how the latent plan maps to the env-action sequence), not the latent representation. This is the same decoding bottleneck that §9.1 (latent-goal MPC) and §6 (env-SR saturates) already established; this targeted probe confirms it on the most trace-friendly task we have.

Honest scope. The trace interface may still be distinguished on tasks where the **decoding bottleneck is removed** — e.g. by a hand-crafted controller that maps the latent directly to a known target state without going through CEM — but such a controller is no longer testing the *predictive-state* question. We do not have such a result, and we do not claim one. The full per-cell JSONs are at `results/generalist_G15_trace_demo/eval/` and the summary table is `results/generalist_G15_trace_demo/eval/RESU`

7.6 Event-Window gating experiment (v0.7.11 protocol, partial result)

To test the content-aware-rate-counter hypothesis from §9.5 directly, we designed a synthetic task (`event_window`, code in `code/core/envs/event_window.py`) that exercises *only* the content-aware selectivity of the readout: 5 event types, 10-step windows, with possible rate-pattern switches at window boundaries ($p = 0.30$). The agent must report the modal event of the current window. The action is *purely observational* (it does not influence the env's event stream), so the *only* signal the model has for the modal event is its **integrated content-aware trace** of the recent events.

Result: STJEWM readouts (trace, membrane) both win over CuBiFAE on this task by ∼2 percentage points, direction- consistent across all 3 seeds. The trace and membrane readouts tie on this task ($p \approx 0.25$).

Interpretation. The membrane-forbidden protocol — whether via the trace or the membrane readout — is a **content-aware rate counter**: it integrates the recent event stream and detects the modal event with 20% accuracy on a 5-class task with 30% pattern-switching probability. CuBiFAE's passive fixed-τ

Model	mean_reward (per 20 windows)	%
`cubifae_baseline`	3.67 ± 0.21	18.4%
`stjewm_trace_only`	4.01 ± 0.16	20.1%
`stjewm_membrane_readout`	4.19 ± 0.11	20.9%

Table 7: Event-Window task result. STJEWM readouts (trace, membrane) both win over CuBiFAE on this content-aware task by ∼2 percentage points, direction-consistent across all 3 seeds. Trace and membrane readouts tie on this task.

decay is strictly less informative on this content-aware dimension. The 50-pp gap to the 70% oracle is the same plan-to-action decoding bottleneck named in §9.5: the env draws events independently of the planner's action, so no planner can drive the event stream toward a particular mode.

Honest scope. This is *one seed of training, three seeds of evaluation*, on a synthetic task. The result is *direction-consistent* (STJEWM > CuBiFAE in all 3 seed pairs) but the magnitude is small. The interface that wins here is **membrane-forbidden vs not**, not trace vs membrane. The trace interface is a *specific instance* of the membrane-forbidden family; on tasks where the membrane readout ties the trace readout (§6, §7, §9.1, §9.5, §9.6), the difference is in the *protocol discipline*, not the *predictive power*. Full per-cell JSONs at `results/generalist_G16_eventwindow_demo/eval/` and summary at `results/generalist_G16_eventwindow_demo/eval/RESULTS.md`.

7.7 Cross-benchmark family OOD (v0.7.12, partial negative result)

The cross-benchmark-family axis is the *true* OOD axis (different benchmark families, not just different sub-families of DMC). We ran 3 splits (one family held out at a time, from the 4-family set {DMC, Reacher, PushT, TwoRoom}):

Split (held-out family)	Model	Eval env	LeWM-SR	env-SR
F1 (PushT)	cubifae_baseline	pusht	0.156	0.000
F1 (PushT)	stjewm_trace_only	pusht	0.233	0.000
F1 (PushT)	**stjewm_membrane_readout**	pusht	**0.400**	0.000
F2 (TwoRoom)	cubifae_baseline	tworoom	**0.878**	0.000
F2 (TwoRoom)	stjewm_trace_only	tworoom	0.778	0.000
F2 (TwoRoom)	stjewm_membrane_readout	tworoom	0.756	0.000
F3 (Reacher)	cubifae_baseline	reacher	0.533	0.033
F3 (Reacher)	stjewm_trace_only	reacher	0.578	0.033
F3 (Reacher)	stjewm_membrane_readout	reacher	0.556	0.033
F4 (DMC held out)	cubifae_baseline	13 DMC envs (avg)	0.506	—
F4 (DMC held out)	stjewm_trace_only	13 DMC envs (avg)	0.506	—
F4 (DMC held out)	stjewm_membrane_readout	13 DMC envs (avg)	0.518	—

Table 8: Cross-benchmark family OOD result. STJEWM does not have a universal hard performance win on cross-benchmark-family transfer. The result is split: STJEWM (membrane readout) wins clearly on F1 (+24.4 pp on LeWM-SR); CuBiFAE wins on F2 (+10 pp); all three tied within noise on F3.

Result: STJEWM does not have a universal hard performance win on cross-benchmark-family OOD. The result is split: STJEWM (membrane readout) wins clearly on F1 / PushT (+24.4 pp on LeWM-SR); CuBiFAE wins on F2 / TwoRoom (+10 pp); all three tied within noise on F3 / Reacher. STJEWM does not generalise uniformly across benchmark families.

Implication for the working title. The working title "generalisable world models" is **supported within DMC sub-families** (v0.7.10b, §7.6) but **not supported across benchmark families** (v0.7.12, this section). The honest scope is that STJEWM is a *methodological contribution* (a probe for the membrane-forbidden protocol), not a *universal-performance contribution* on cross-benchmark-family transfer. Full per-cell JSONs at `results/cross_benchmark_F{1,2,3}/eval/` and summary at `results/cross_benchmark_F1/eval/RESULTS.`

Table 1 — Main claim control table

Claim	Status
STJEWM competitive on env-native success	**suppo**
STJEWM raw SOTA	**not c**
LeWM-SR alone is insufficient; can be inflated	**suppo**
STJEWM latents are event-aligned (specialist + generalist)	**suppo**
STJEWM family lands in the calibrated regime under shared weights	**suppo**
Three non-spiking failure modes (collapse / noise / over-reactive) are diagnosable from a 200-step random-policy trajectory	**suppo**
Membrane-readout catastrophically fails stress	**refute**
Planner causally relies on the event-window trace component specifically	**refute**
STJEWM generalist stress env-SR	**not c**
Multi-seed std bars on generalist eval	**defer**
Membrane-forbidden protocol is empirically necessary on stress	**not c**

Table 9: **Reading guide.** Rows marked **supported** are the paper's central empirical findings; the paper would not be retracted if any one of them were weakened. Rows marked **refuted** are claims that *used* to be in the paper and were dropped on re-evaluation; their refutation is part of the audit trail. Rows marked **not claimed** are explicitly outside the paper's scope. Rows marked **deferred** are honest placeholders for work the paper's design would have caught but that the wallclock budget did not permit. Per-env matrices and per-suite raw G4 / G8 / G16 numbers live in `MASTER_TABLE.md` ğ1–ğ9.7, the operational source of truth.

Appendix outline

Appendix A — Implementation details. Environment list (16 standard + 4 stress + LeWM), training hyperparameters, CEM planner settings, all six readout-mode implementations, data generation pipeline (multi-env spec format).

Appendix B — Specialist per-env tables. All 20 standard envs × 13 models per metric (env-SR, LeWM-SR); all 4 stress envs.

Appendix C — Generalist per-suite tables. G4 / G8 / G16 raw matrices for env-SR, LeWM-SR, divergence, responsiveness, event-align ρ per (env, model) cell.

Appendix D — Event probe details. Event labels, AUROC per env-target, probe training protocol, hold-out splits.

Appendix E — Event alignment details. Event-boundary definition (observation first-difference local maxima), Pearson ρ per env, computation protocol.

Appendix F — Collapse-diagnostic derivation. Toy proof that a constant latent has div = 0 regardless of planner. Responsiveness definition with bounded-gain caveat. Alignment-vs-divergence independence argument.

Appendix G — Historical experiments and claim revision audit. v0.4 "membrane collapses under stress" → v0.7.2 refutation. v0.7.4 "LeWM-SR collapse signature invisible" → v0.7.5 diagnostic. v0.7.3 pilot ("4-env subset too small") → v0.7.5 16-env generalist. This appendix makes the empirical story self-auditing rather than history-of-projects.

References

[1] LeWM-style joint-embedding world model (Hafner et al., 2023; this paper's repo, `README.md`).

[2] CuBiFAE (Kaiser et al., 2024 ICML).

[3] SpikeDreamer (Hong et al., 2024 AAAI).

[4] SLT-LIF (Liu et al., 2024 NeurIPS workshop).

[5] STJEWM `ReadoutMode` enumeration, `code/stjewm.py` line 50 (`RATE_ONLY` docstring: "moving_average(spike).detach()").